How model_6999 was built: from static clutter to a remembering local planner

Dynamoid wheeled humanoid · design journey, August – October 2026 · every number below is taken from the project's own records (appendix)

1. The short version

The final policy (`MEM_GRU_model_6999`) is the fourth design, not the first. Each earlier design failed in a way that told us what the next one needed.

era	what we built	what it taught us
1. Static clutter (Aug)	RL policy on BARN-style box courses, lidar input	static traversal plateaued at ~46% success; the real target was *dynamic* avoidance
2. Privileged teacher (Aug–early Sep)	PPO teacher that sees pedestrians as a time-to-collision (TTC) scan → `model_699`	TTC is the right language for risk; but a teacher that sees more than the camera cannot be distilled into a camera student
3. Rebuild (mid Sep)	policy = local planner inside a global trajectory, end-to-end from depth, asymmetric critic → `SOLID_BASE_4499`	the formulation and the reward arithmetic were as wrong as the architecture; coverage is not the whole problem
4. Smooth, then remember (late Sep)	action-rate penalty → `5499`; GRU memory → `model_6999`	pedestrians leave the 120° view just before impact; memory halves the hits

2. Era 1: static obstacles (BARN-style)

The first task was a traversal benchmark rebuilt at this robot's scale: a walled 16 m × 11 m arena, a fixed start and goal, up to 20 boxes, 1,000 frozen courses in four difficulty levels (easy to hard, difficulty *measured* as the widest-path bottleneck clearance, not requested). Scoring followed the BARN challenge: `s = success × OT / clip(AT, 2·OT, 8·OT)`, where OT is the optimal traversal time, so the maximum is 0.5 (the 2026 BARN leader scored 0.4975 on a different robot). The robot's safety disc was 0.791 m, set by its outstretched hands.

The RL baseline reached a BARN score of 0.2268 and 46.2% success (4.2% collisions); on the hard level it solved 18%. Relaxing the curriculum gate unlocked the hard level but did not raise the overall result (44.6% success, 0.2159). The team already had a static-obstacle pipeline, and the project's stated end goal was static *and dynamic* avoidance, so the work moved to people. From BARN's dynamic sibling, **DynaBARN**, we took one design rule: benchmarks that use a single motion model (constant velocity, social force, ORCA) breed policies that beat that model and fail on anything else, so motion profiles should be varied.

3. Era 2: a privileged teacher, and `model_699`

3.1 Making the problem reachable

The first pedestrian task gave the robot a fixed goal 8 m away. For nine million training steps it reached the goal zero times: to find the +100 at the end it had to chain about 190 forward steps through an encounter that ends at −100. The fix was cheap: sample the start position so that *some* episodes are winnable in two seconds, and the goal enters the value function and propagates outward (a reverse curriculum). Most of what followed was making problems reachable and measurements honest, not making policies cleverer.

The task was also migrated from the prototype (v2) to the machine actually built (v1), which had to be re-characterised: a 0.6067 m safety disc (not v2's 0.791 m), castors that are fixed

skids modelled as spheres, a drive that delivers 1.37 m/s against the real robot's 1.6, and a root frame that points along +y, so every "forward speed" read from column 0 returned about zero on a robot doing 1.37 m/s. None of these raised an error.

3.2 The observation that mattered: a TTC scan

At Stage 2 the task became a *local planner*: track a waypoint 2–3 m ahead and avoid pedestrians, with routing left to a global planner. Avoidance levers had bought almost nothing at 8 m because the residual failures were routing failures. Then the observation changed. The policy had been given twenty pedestrian slots sorted by threat; navigation needs the *space* between people, and sorting by threat scatters the two pedestrians bounding a gap into non-adjacent slots. Replacing the list with a directional **time-to-collision scan** (24 bearings, "when would I hit something if I drove this way") gave the largest single gain of the project:

4 pedestrians	slot list	TTC scan
arrival	0.551	**0.828**
collision	0.422	**0.109**
median clearance	0.155 m	**0.604 m**

3.3 What was good about `model_699`

`model_699` (Stage 5, trained 2026-08-29) is the open-arena teacher: 8 pedestrians, 25 s episodes, **91.5% arrival and 8.5% collision (n = 1,024)**. It saw a 32-number observation (the 24-bearing, 200°, 3 s TTC scan, its own twist, a goal vector clipped to 3 m, and its last action) through a tiny MLP (256-128-64-2), trained in 700 iterations on 2,048 environments. Its strengths:

- The scan is a good state: it is image-shaped, reads free space directly, and made collisions four times rarer than the slot list.
- It is cheap and reproducible (self-contained archive with config and weights), and it is a strong **ceiling**: it shows what a policy can do when perception is perfect.
- Its design set the reward economics we still use: progress that telescopes (so it cannot be farmed) and a dense-cost budget small enough that arriving always beats not arriving.
- Its pedestrians never yield, so its numbers are conservative.

3.4 Where the teacher line ran out

1. **A 200° arc nobody derived.** It was chosen as "wider than the camera", written down as a constant, and then a document claimed it matched the Femto camera. It never did. A placeholder became a spec.
2. **Coverage privilege does not distil.** A teacher that reacts to pedestrians behind it labels identical camera frames with conflicting actions. A policy whose scan was restricted to a 90.8° arc scored 0.536 collisions at 8 pedestrians against 0.101 for the 200° teacher, and adding one second of frame stacking made it *worse* (0.568): memory only pays if the robot goes and looks, and nothing rewarded information. (Later work showed 90.8° itself was an artifact: the real camera is 120°, and its range is 2.88 m, not the 20 m the simulator had been using.)
3. **A camera-matched teacher helped, but not enough.** Stage 12, trained in-cone, reached 0.891 arrival / 0.092 collision at 4 pedestrians. Moved into a warehouse aisle, the best teacher (`model_1962`) stalled at 0.621 / 0.242, with 96.8% of its collisions arriving from outside the cone. Five attempts to add memory in this framework did not beat it, and three of our own analyses (including a planner-based "ceiling") were retracted when the instrument that produced them failed its check.

4. **The pricing of failure.** Success +100, collision −100 and timeout −100 priced giving up and hitting someone identically (break-even collision risk 0.75). It did not stop 699 in the open arena, but when harder tasks stalled with collisions stuck near 0.6 for thousands of iterations, the arithmetic was the first suspect.
5. **No global path.** Every run before mid-September trained a *navigator* with a single goal pose, so "wandering" had no reference to be measured against.

4. The pivot (mid September)

On 2026-09-17 the formulation was rewritten from first principles (design record `SOCIAL_NAV_DESIGN.md`):

- **The policy is a local planner.** The sim owns a global trajectory; the policy tracks it, avoids people, and rejoins. This is the same split Nav2 uses (global planner → path → controller), and the DRL-VO line of work, which this project kept comparing against, also puts its policy under a global planner.
- **The environment was exonerated.** A probe swept the real action space for every conflicting pedestrian pair and found no pair with no avoiding control (0 of 1,140; 1.1% were blind shoulder grazes).
- **The reward was fixed.** Timeout moved from −100 to −600 so patience is not priced like a crash; a budget rule keeps all dense costs combined below the collision penalty; tracking and schedule terms were added so a dodge has a line to rejoin. A guard written to enforce that budget turned out to grade its own hard-coded table, not the config, and passed a reward that inverted the economics.
- **The sensor was locked to measurement.** 120° × 120°, 0.25–2.88 m, mounted flat 0.09 m up and 0.40 m ahead. Two long-held "facts" (90.8° and 20 m) were artifacts, and an earlier mount had the camera facing backward.
- **Teacher–student was dropped for end-to-end PPO from depth with an asymmetric critic.** A privileged teacher reacts to pedestrians outside the camera's field of view, which a depth-only student structurally cannot reproduce at any level of training. So the actor sees only what the robot has; the critic still sees privileged state (including the 24-direction TTC scan first built for the teacher) and exists only in training.

The first trusted base was the **treadmill corridor** policy `SOLID_BASE_model_4499` (3 pedestrians, goal on the path, box-free): 69.4% goal, 28.2% hit in the later 507-episode evaluation.

5. Static obstacles are slow dynamic obstacles

Static avoidance did not need its own module. The actor sees a 4 × 84 × 84 depth stack with no class label: a box and a pedestrian are both pixels at a range, and a static box is a pedestrian whose velocity is zero. Zero velocity *removes* the prediction problem rather than adding one, and the boxes are larger than the pedestrians, so the depth signal is stronger. The check: `model_2999`, which had never seen a box in training, hit boxes in only **4.3%** of 322 deterministic episodes. The project owner's framing at the time was "a static box is nothing but a moving box with zero velocity". The one genuine difference is not static versus dynamic but *one obstacle in frame versus many*: boxes in the corridor compound pedestrian collisions by eating dodge room (two-pedestrian collision 0.47–0.58 with boxes versus 0.256 without), so the base policy trains box-free. By the same logic, a remembered static obstacle is the v = 0 case of the state the memory is designed to carry (not separately tested).

6. From 4499 to 6999

Smoothness (4499 → 5499). The policy was jerky: the action-rate penalty was not seeing the problem because it acted on raw actions that were seven times past the clamp. Penalising the *clamped* action rate removed 5 Hz white-noise chatter; goal arrival rose to 75.8% and hits fell to 22.7%.

Memory (5499 → 6999). The first diagnosis ("100% of hits happen outside the camera cone, blind for 3.9 s") was wrong: a re-measurement with a frame check (143 hits) found 99.3% of hits were pedestrians the camera *had seen, then lost*, blind for a median 0.3 s. They leave the cone at about 1.2 m, beside the robot, and 31% of those exits are caused by the robot's own turning. The four-frame stack holds only 0.2 s, so 58% of hits happen after more than 0.2 s blind and need memory.

- *Why a GRU.* The pedestrians move at constant velocity, so each pedestrian's relative position and velocity is a sufficient statistic and its update is a near-linear recurrence driven by the robot's own twist (a Kalman filter is a recurrence). The state is tiny, so the cell type is not the bottleneck: GRU over LSTM for one tensor to carry in the C++ stack and about 25% fewer parameters.
- *Two things decide whether memory is learned.* (1) The back-propagation horizon: rsl_rl only back-propagates within one rollout, so rollouts went from 24 to 48 steps (2.4 s, covering about 91% of measured blind times). (2) The signal: a collision penalty arrives seconds later in about a quarter of episodes, so a dense auxiliary loss makes the GRU state predict, per 15° bearing sector, the proximity of pedestrians seen earlier.
- *Safe warm start.* The memory joins the head through zero-initialised columns, so the new model began identical to its parent; carrying Adam's state across stages mattered (resetting it dropped goal arrival from 75% to 8% in ten iterations).

Result: hits 22.7% → 11.3%, goal arrival 75.8% → 86.9%.

7. Architecture choices and where they came from

choice	source / reasoning	outcome
asymmetric actor–critic PPO (rsl_rl, Isaac Lab)	deployable actor sees only robot sensors; the critic's privileged state speeds learning	kept
conv stack 32@8×8/4, 64@4×4/2, 64@3×3/1	the standard Atari/DQN visual stack, on 84 × 84 frames	kept; it holds only 78 k of 2.9 M parameters
MLP head 512-256-128, ELU	legged_gym / rsl_rl defaults ("a standard recipe, not a search result")	kept
TTC cost and TTC scan	own formulation; the scan became the critic's privileged input	kept
goal-radius ramp (1.5 → 0.3 m in Arena)	the idea, not the numbers: arrival radius is a reachability knob, set to 1.0 m here	adopted
personal space 0.5 m surface-to-surface	copied from **Arena** (`ped_type_safety_distance`) so the metric matches Arena's	kept
Arena's softer person-than-wall penalty and fixed dense:terminal ratio	copied in Phase 2, **deliberately not reproduced**: the real mechanism is variance, and the ratio is superseded by the budget bound	dropped
DRL-VO as comparison	local planner under a global one; its pedestrians yield, ours never do	reference only
MPC at the core	rejected: needs a tracker the actor does not have, plant error is ~11%, avoidance is non-convex; kept only MPC's constraint discipline (the wall-stop reflex)	rejected
explicit pure-pursuit hand-off	more sample-efficient, but adds a mode boundary; RL stays in control and the tracking cost teaches rejoining	not chosen for training
HIL-SERL	policy improvement, not perception	not applicable
bounded action mean	pulled the mean in range; goal arrival fell from 75% to 1–28%	rejected

8. `model_699` and `model_6999` side by side

	`model_699` **(Stage 5)**	`model_6999`
role	privileged **teacher**	**local planner** intended for the robot
world	open arena, no walls or path, 8 pedestrians	walled corridor, 10 m path, 3 pedestrians
sensing	analytic TTC scan from ground truth, 200°, perfect	4 depth frames, 120°, 2.88 m, plus 128-number memory
task	reach a goal (clipped to 3 m)	follow a global path, avoid, rejoin
actor input	32 numbers	28,224 + 9 + 128
network	MLP 32-256-128-64-2	CNN + GRU + MLP, 2,875,452 parameters
reward	goal +100, collision −100, timeout −100	goal +100, hit −200, timeout −30, TTC −4, personal −14, track/schedule +1, action rate −0.5
result	91.5% goal, 8.5% hit (n = 1,024)	86.9% goal, 11.3% hit (n = 929)
runs on the robot?	no (needs the lidar scan)	not yet (needs ONNX with GRU state)

The two results are **not comparable**: 699 sees every pedestrian perfectly in an open arena, while 6999 sees only a 120° cone out to 2.88 m in a corridor. That a depth-only policy lands within a few points of a perfect-information teacher is the point of the redesign.

Carried over from 699: TTC as the language of risk; the reverse-curriculum trick; the local-planner interface (a waypoint, now a path point 1.2 m ahead); the reward economics; non-yielding pedestrians as a pessimistic test; and the rule that a sensor arc must be measured, never assumed.

9. What failed, and what worked

failed or retracted	worked
fixed 8 m goal (nine million steps, zero arrivals)	start sampling so some episodes are winnable
pedestrian slot list	directional TTC scan
200° "camera-matched" arc	measuring the real mount and field of view
coverage-privileged teacher → student	end-to-end depth actor with privileged critic only
reactive-pedestrian curriculum (looked best in training logs, 94.9% wins; worst at the pessimistic rate)	judging at the non-reactive rate; pooling runs
four reward/architecture theories, killed in one day by measurement	looking at where hits actually occur
"100% of hits outside the cone, blind 3.9 s"	re-measuring with a frame check
three aisle analyses, including a DWA "ceiling" (retracted)	validating every instrument against a trusted number
longer memory window; CNN trained by the memory loss; discrete speed levels	GRU side branch + auxiliary sector loss + Adam carry-over

10. Honest limits

Simulation only: a straight corridor and three non-reactive pedestrians. The policy steers but does not choose to slow or stop (raw speed is beyond the clamp on 99.6% of steps), and hits after more than 1.2 s out of view are now 47% of what remains. Deployment needs the ONNX export with GRU state, a C++ node, and an independent 360° lidar safety layer with a speed cap.

Appendix: where each claim lives

All paths under `wheeled_humanoid_manager/`. Era 1: `AGENT_INSTRUCTIONS.md`, `reports/M0_*.md`, `reports/M1_*.md`. Era 2: `rl/dynamoid/WRITEUP.md`, `rl/dynamoid/STAGES.md`, `rl/figs/checkpoints/STAGE5_SCAN_model_699/README.md`, `reports/AISLE_*.md`. Pivot: `rl/SOCIAL_NAV_DESIGN.md`, `rl/AXES_AND_MOUNT.md`. Memory: `rl/dynamoid/memory_policy.py`, `rl/figs/checkpoints/MEM_GRU_model_6999.md` and `_eval_REPORT.md`. Figures and the model report: `rl/figs/report/MEM_GRU_model_6999/REPORT.md`.